Lab 3 - Newton's and Fixed Point Methods

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Problem 1

```
function [result] = MyNewtonMethod(x1,tol,Nmax,f,fprime)
arguments
    x1 double
    tol double
    Nmax double
    f function_handle
    fprime function_handle
end
    xs = [x1];
    ys = [f(x1)];
    xn = x1
    i = 1;
    while abs(f(xn)) > tol && i < Nmax
       xn = x1 - ((f(x1))/(fprime(x1)))
       xs = [xs; xn];
       ys = [ys; f(xn)];
       x1 = xn
       i = i + 1;
    end
    cols = {'x';'y'};
    xn = xs
    yn = ys
    T = table(xn,yn);
    result = T;
end
```

Doing it for the specified function, we get the following table

```
5×2 table

      xn          yn
    ______    __________

         2            -3
      2.75        0.5625
    2.6477       0.01046
    2.6458    3.9015e-06
    2.6458    5.4357e-13
```

The built-in sqrt() function in matlab yields the exact same x value

Problem 2

```
function [result] = MyNewtonMethod(x1,tol,Nmax,f,fprime)
```

```
arguments
    x1 double
    tol double
    Nmax double
    f function_handle
    fprime function_handle
end
    xs = [x1];
    ys = [f(x1)];
    xn = x1;
    En1 = [0; 0];
    En2 = [0; 0];
    En3 = [0; 0];
    i = 1;
    while abs(f(xn)) > tol && i < Nmax

        xn = x1 - ((f(x1))/(fprime(x1)));
        xs = [xs; xn];
        ys = [ys; f(xn)];
        if i > 1
            En1 = [En1; (abs(x1-xn))/(en)];
            En2 = [En2; (abs(x1-xn))/(en^2)];
            En3 = [En3; (abs(x1-xn))/(en^3)];
        end
        en = abs(x1-xn);
        x1 = xn;
        i = i + 1;
    end
    cols = {'x';'y';'En1';'En2';'En3'};
    xn = xs
    yn = ys
    T = table(xn,yn,En1,En2,En3);
    result = T;
end
```

Running it yields the following table

10×5 table

xn	yn	En1	En2	En3
1	0.037037	0	0	0
0.88889	0.010974	0	0	0
0.81481	0.0032515	0.66667	6	54
0.76543	0.00096342	0.66667	9	121.5
0.73251	0.00028546	0.66667	13.5	273.38
0.71056	8.458e-05	0.66667	20.25	615.09
0.69593	2.5061e-05	0.66667	30.375	1384
0.68618	7.4254e-06	0.66667	45.562	3113.9

```
    0.67967    2.2001e-06    0.66667    68.344    7006.3
    0.67534    6.5189e-07    0.66667    102.52     15764
```

Since the linear error is constant, this converges linearly. This is because the function's derivative is 0 at the root.

Problem 3

```
function [result] = MyFixedPoint(x1,tol,Nmax,g)
arguments
    x1 double
    tol double
    Nmax double
    g function_handle
end
    xs = [x1];
    ys = [g(x1)];
    xn = x1;
    En1 = [0; 0];
    En2 = [0; 0];
    En3 = [0; 0];
    i = 1;
    while abs(g(xn)-xn) > tol && i < Nmax

        xn = g(x1)
        xs = [xs; xn];
        ys = [ys; g(xn)];
        if i > 1
            En1 = [En1; (abs(x1-xn))/(en)];
            En2 = [En2; (abs(x1-xn))/(en^2)];
            En3 = [En3; (abs(x1-xn))/(en^3)];
        end
        en = abs(x1-xn);
        x1 = xn;
        i = i + 1;
    end
    cols = {'x';'y';'En1';'En2';'En3'};
    xn = xs
    yn = ys
    T = table(xn,yn,En1,En2,En3);
    result = T;
end
```

Problem 4

(a)

The result from Newton's method:

```
5×5 table
```

xn	yn	En1	En2	En3
1.5	2.375	0	0	0
1.3733	0.13435	0	0	0
1.3653	0.00052846	0.063721	0.50306	3.9715
1.3652	8.2905e-09	0.0039648	0.49122	60.86
1.3652	0	1.5689e-05	0.49025	15320

(b)

i.

12×5 table

xn	yn	En1	En2	En3
1.5	1.3484	0	0	0
1.3484	1.3674	0	0	0
1.3674	1.365	0.12518	0.82569	5.4465
1.365	1.3653	0.12749	6.7183	354.03
1.3653	1.3652	0.1272	52.574	21731
1.3652	1.3652	0.12723	413.46	1.3436e+06
1.3652	1.3652	0.12723	3249.5	8.2992e+07
1.3652	1.3652	0.12723	25540	5.127e+09
1.3652	1.3652	0.12723	2.0074e+05	3.1673e+11
1.3652	1.3652	0.12723	1.5778e+06	1.9567e+13
1.3652	1.3652	0.12723	1.2401e+07	1.2088e+15
1.3652	1.3652	0.12723	9.7471e+07	7.4673e+16

ii.

5×5 table

xn	yn	En1	En2	En3
1.5	1.3733	0	0	0
1.3733	1.3653	0	0	0
1.3653	1.3652	0.063721	0.50306	3.9715
1.3652	1.3652	0.0039648	0.49122	60.86
1.3652	1.3652	1.5689e-05	0.49025	15320

iii.

20×5 table

xn	yn	En1	En2	En3
1.5	1.287	0	0	0

1.287	1.4025	0	0	0
1.4025	1.3455	0.54254	2.5466	11.953
1.3455	1.3752	0.49385	4.2725	36.964
1.3752	1.3601	0.52051	9.1185	159.74
1.3601	1.3678	0.50741	17.078	574.77
1.3678	1.3639	0.51424	34.11	2262.5
1.3639	1.3659	0.51078	65.884	8498.1
1.3659	1.3649	0.51256	129.44	32686
1.3649	1.3654	0.51165	252.08	1.2419e+05
1.3654	1.3651	0.51212	493.13	4.7484e+05
1.3651	1.3653	0.51188	962.46	1.8097e+06
1.3653	1.3652	0.512	1880.7	6.9082e+06
1.3652	1.3652	0.51194	3672.8	2.6349e+07
1.3652	1.3652	0.51197	7174.7	1.0054e+08
1.3652	1.3652	0.51196	14013	3.8358e+08
1.3652	1.3652	0.51196	27373	1.4635e+09
1.3652	1.3652	0.51196	53465	5.5836e+09
1.3652	1.3652	0.51196	1.0443e+05	2.1303e+10
1.3652	1.3652	0.51196	2.0399e+05	8.1277e+10

iv.

8×5 table

xn	yn	En1	En2	En3
1.5	3.875	0	0	0
3.875	112.12	0	0	0
112.12	1.46e+06	45.578	19.191	8.0803
1.46e+06	3.1118e+18	13486	124.58	1.1509
3.1118e+18	3.0134e+55	2.1316e+12	1.4602e+06	1.0002
3.0134e+55	2.7362e+166	9.6835e+36	3.1118e+18	1
2.7362e+166	Inf	9.0803e+110	3.0134e+55	1
Inf	Inf	Inf	NaN	NaN

v.

5×5 table

xn	yn	En1	En2	En3
2	1.4107	0	0	0
1.4107	1.3653	0	0	0
1.3653	1.3652	0.077123	0.13088	0.22209
1.3652	1.3652	0.00080712	0.017759	0.39076
1.3652	1.3652	5.6296e-10	1.5347e-05	0.41839

g1:

$$x = \sqrt{\frac{10}{4+x}}$$
$$x^2 = \frac{10}{4+x}$$
$$x^2(4+x) = 10$$
$$x^3 + 4x^2 - 10 = 0$$

g2 is just Newton's method

g3:

$$x = \sqrt{\frac{10-x^3}{4}}$$
$$x^2 = \frac{10-x^3}{4}$$
$$4x^2 = 10 - x^3$$
$$x^3 + 4x^2 - 10 = 0$$

g4:

$$x = x^3 + 4x^2 + x - 10$$
$$x^3 + 4x^2 - 10 = 0$$

(c)

I already made the tables prior

Analysis:

g1 and g3 are linear, g2 is quadratic, g4 doesn't converge, and g5 is maybe cubic in convergence. The linear ones are as such because g'(x*) isn't zero, the quadratic one is because it is zero, the cubic one is because it and g''(x*) are zero, and g4 not converging is because |g'(x*)| is greater than one

This shows that Newton's method, when it works to its strengths, is really really really good. The only thing as good or better were the fixed point iterations that were just newton's method (g2) and newton's method but better (g5)

Problem 5

Our equation is as follows:

$$Cone = \frac{1}{3}\pi r^2 h$$

$$Semisphere = \frac{2}{3}\pi r^3$$

$$\frac{10}{3}\pi r^2 + \frac{2}{3}\pi r^3 = 60$$

Bisection:

```
21×2 table
```

xn	yn
1	-47.434
10	3081.6
5.5	605.23
3.25	122.51
2.125	7.3847
1.5625	-26.444
1.8438	-11.274
1.9844	-2.3985
2.0547	2.3775
2.0195	-0.039136
2.0371	1.162
2.0283	0.55963
2.0239	0.2598
2.0217	0.11022
2.0206	0.035514
2.0201	-0.001818
2.0204	0.016846
2.0202	0.0075138
2.0201	0.0028478
2.0201	0.00051489
2.0201	-0.00065155

Secant:

10×2 table

xn	yn
1	-47.434
10	3081.6
1.1364	-43.402
1.2595	-39.202
2.4086	30.019
1.9103	-7.1848
2.0065	-0.91784
2.0206	0.035622
2.0201	-0.00016531
2.0201	-2.9534e-08

Regula Falsi:

86×2 table

xn	yn

```
         1        -47.434
        10         3081.6
    1.1364        -43.402
    1.2595        -39.202
    1.3693        -34.987
    1.4662        -30.886
    1.5509        -26.999
    1.6243        -23.397
    1.6874        -20.121
    1.7413        -17.189
    1.7871          -14.6
    1.8259        -12.341
    1.8585        -10.387
    1.8858        -8.7127
    1.9087        -7.2862
    1.9278        -6.0782
    1.9437        -5.0598
    1.9569        -4.2048
    1.9678        -3.4892
    1.9769        -2.8919
    1.9844        -2.3944
    1.9907        -1.9809
    1.9958        -1.6377
    2.0001        -1.3531
    2.0036        -1.1175
    2.0065       -0.92256
    2.0089       -0.76138
    2.0108       -0.62819
    2.0125       -0.51818
    2.0138       -0.42737
    2.0149       -0.35241
    2.0158       -0.29057
    2.0166       -0.23956
    2.0172       -0.19748
    2.0177       -0.16279
    2.0181       -0.13418
    2.0185       -0.11059
    2.0188      -0.091149
     2.019      -0.075122
    2.0192      -0.061911
    2.0194      -0.051022
    2.0195      -0.042048
    2.0196      -0.034652
    2.0197      -0.028556
    2.0198      -0.023533
    2.0198      -0.019393
    2.0199      -0.015981
    2.0199      -0.013169
    2.0199      -0.010852
```

```
      2.02     -0.0089428
      2.02     -0.0073694
      2.02     -0.0060727
      2.02     -0.0050042
      2.02     -0.0041237
    2.0201     -0.0033982
    2.0201     -0.0028002
    2.0201     -0.0023075
    2.0201     -0.0019015
    2.0201     -0.0015669
    2.0201     -0.0012912
    2.0201      -0.001064
    2.0201     -0.0008768
    2.0201    -0.00072252
    2.0201    -0.00059539
    2.0201    -0.00049062
    2.0201    -0.00040429
    2.0201    -0.00033316
    2.0201    -0.00027453
    2.0201    -0.00022623
    2.0201    -0.00018642
    2.0201    -0.00015362
    2.0201    -0.00012659
    2.0201    -0.00010431
    2.0201     -8.596e-05
    2.0201    -7.0834e-05
    2.0201     -5.837e-05
    2.0201      -4.81e-05
    2.0201    -3.9636e-05
    2.0201    -3.2662e-05
    2.0201    -2.6915e-05
    2.0201    -2.2179e-05
    2.0201    -1.8276e-05
    2.0201     -1.506e-05
    2.0201     -1.241e-05
    2.0201    -1.0227e-05
    2.0201    -8.4273e-06
```

Newton:

```
6×2 table

      xn          yn
    ______    __________

         1       -47.434
    2.7421        61.927
    2.1505        9.2617
    2.0256        0.3706
    2.0201    0.00068467
```

```
    2.0201    2.3518e-09
```

Extra Credit

New function:

$$0 = x^3 + 4x^2 - 10$$
$$-x^3 = 4x^2 - 10$$
$$x^3 = -4x^2 + 10$$
$$3 \cdot \ln(x) = \ln(-4x^2 + 10)$$
$$ln(x) = \frac{\ln(-4x^2 + 10)}{3}$$
$$x = e^{(\frac{\ln(-4x^2+10)}{3})}$$
$$x = \left(e^{\ln(-4x^2+10)}\right)^{\frac{1}{3}}$$
$$x = \sqrt[3]{-4x^2 + 10}$$

100×5 table

xn	yn	En1	En2	En3
2	-1.8171	0	0	0
-1.8171	-1.4748	0	0	0
-1.4748	1.0914	0.089682	0.023495	0.0061551
1.0914	1.7364	7.4963	21.898	63.969
1.7364	-1.2725	0.25137	0.097956	0.038172
-1.2725	1.5215	4.6647	7.2314	11.21
1.5215	0.90435	0.92859	0.30861	0.10256
0.90435	1.8879	0.22089	0.079056	0.028294
1.8879	-1.6206	1.5936	2.582	4.1834
-1.6206	-0.79663	3.5673	3.627	3.6878
-0.79663	1.9541	0.23486	0.066939	0.019079
1.9541	-1.7406	3.3383	4.0514	4.9168
-1.7406	-1.2845	1.3432	0.48831	0.17752
-1.2845	1.5038	0.12346	0.033416	0.0090444
1.5038	0.98463	6.1124	13.4	29.374
0.98463	1.8294	0.18619	0.066777	0.023949
1.8294	-1.5016	1.6271	3.1343	6.0373
-1.5016	0.99336	3.9433	4.6682	5.5263
0.99336	1.8225	0.74903	0.22486	0.067507
1.8225	-1.4866	0.3323	0.13319	0.053381
-1.4866	1.0508	3.9912	4.8139	5.8062
1.0508	1.7741	0.76679	0.23173	0.070028
1.7741	-1.3732	0.28507	0.11235	0.044278
-1.3732	1.3495	4.3512	6.0156	8.3167
1.3495	1.3952	0.86508	0.27487	0.087336

1.3952	1.3033	0.016803	0.0061717	0.0022668
1.3033	1.4745	2.0095	43.923	960.08
1.4745	1.0923	1.8628	20.263	220.41
1.0923	1.7356	2.2321	13.034	76.111
1.7356	-1.2701	1.6829	4.4025	11.517
-1.2701	1.5252	4.6722	7.263	11.29
1.5252	0.88599	0.93	0.30942	0.10295
0.88599	1.9001	0.22867	0.081806	0.029266
1.9001	-1.6438	1.5866	2.4822	3.8835
-1.6438	-0.93144	3.4946	3.4459	3.398
-0.93144	1.8691	0.20101	0.05672	0.016005
1.8691	-1.584	3.9314	5.519	7.7476
-1.584	-0.32941	1.233	0.44027	0.15721
-0.32941	2.1228	0.36332	0.10522	0.03047
2.1228	-2.0021	1.9546	1.558	1.2419
-2.0021	-1.8205	1.6821	0.68596	0.27973
-1.8205	-1.4823	0.044024	0.010673	0.0025874
-1.4823	1.066	1.8624	10.256	56.477
1.066	1.7603	7.5346	22.278	65.871
1.7603	-1.3379	0.27249	0.10693	0.041962
-1.3379	1.4161	4.462	6.4261	9.2547
1.4161	1.2555	0.88889	0.2869	0.092599
1.2555	1.546	0.05831	0.021173	0.0076879
1.546	0.76043	1.809	11.265	70.149
0.76043	1.9736	2.7042	9.3087	32.044
1.9736	-1.7737	1.5443	1.9659	2.5025
-1.7737	-1.3722	3.0889	2.5462	2.0988
-1.3722	1.3514	0.10714	0.028591	0.00763
1.3514	1.3915	6.784	16.898	42.089
1.3915	1.3113	0.014719	0.0054041	0.0019841
1.3113	1.4616	2.002	49.94	1245.7
1.4616	1.1332	1.873	23.337	290.78
1.1332	1.6943	2.1849	14.535	96.692
1.6943	-1.1403	1.7086	5.2023	15.84
-1.1403	1.6867	5.0515	9.0021	16.042
1.6867	-1.1134	0.99733	0.35184	0.12412
-1.1134	1.7146	0.99051	0.35037	0.12394
1.7146	-1.2073	1.01	0.36067	0.1288
-1.2073	1.6095	1.0332	0.36533	0.12918
1.6095	-0.71285	0.96403	0.32993	0.11291
-0.71285	1.9973	0.82446	0.29269	0.10391
1.9973	-1.8127	1.167	0.50249	0.21637
-1.8127	-1.4649	1.4058	0.51873	0.19141
-1.4649	1.1229	0.091284	0.023959	0.0062885
1.1229	1.705	7.4409	21.395	61.516
1.705	-1.1763	0.22491	0.086908	0.033583
-1.1763	1.6467	4.9504	8.5054	14.613
1.6467	-0.94615	0.97979	0.34006	0.11803
-0.94615	1.8585	0.91848	0.32536	0.11525
1.8585	-1.5627	1.0817	0.41717	0.16089

```
     -1.5627     0.61467      1.2198     0.43493     0.15508
     0.61467      2.0399     0.63643     0.18603    0.054376
      2.0399       -1.88     0.65459     0.30064     0.13808
       -1.88     -1.6055      2.7504      1.9297      1.3539
     -1.6055    -0.67712     0.07004    0.017867    0.004558
    -0.67712      2.0137      3.3814      12.316      44.858
      2.0137     -1.8391      2.8985      3.1221       3.363
     -1.8391     -1.5225      1.4318     0.53211     0.19775
     -1.5225     0.89933    0.082166    0.021326   0.0055351
     0.89933      1.8913      7.6503      24.166      76.335
      1.8913     -1.6271     0.40957     0.16911    0.069827
     -1.6271    -0.83852       3.547      3.5758      3.6049
    -0.83852      1.9299     0.22413    0.063703    0.018106
      1.9299     -1.6982      3.5107       4.452      5.6457
     -1.6982     -1.1537      1.3105      0.4734       0.171
     -1.1537      1.6722     0.15008    0.041365    0.011401
      1.6722     -1.0581        5.19      9.5319      17.506
     -1.0581      1.7675     0.96617      0.3419     0.12099
      1.7675     -1.3565      1.0349     0.37905     0.13883
     -1.3565      1.3821      1.1056     0.39128     0.13848
      1.3821      1.3313     0.87663     0.28062    0.089827
      1.3313      1.4278    0.018542   0.0067709   0.0024724
      1.4278      1.2267      1.9002      37.421      736.94
      1.2267      1.5849      2.0844      21.602      223.88
      1.5849    -0.36308      1.7813      8.8564      44.033
```

It does not converge, and interestingly enough, is instead a highly chaotic system rather than simply oscillating between two points or diverging to infinity. When tried with 100,000 steps and a tolerence of 10^-3 we actually do find the root! Perhaps, with enough runtime this will eventually end up finding the root with a better tolerence.